Shopzilla Customer Satisfaction Prediction Using Deep Learning ANN

Project Report

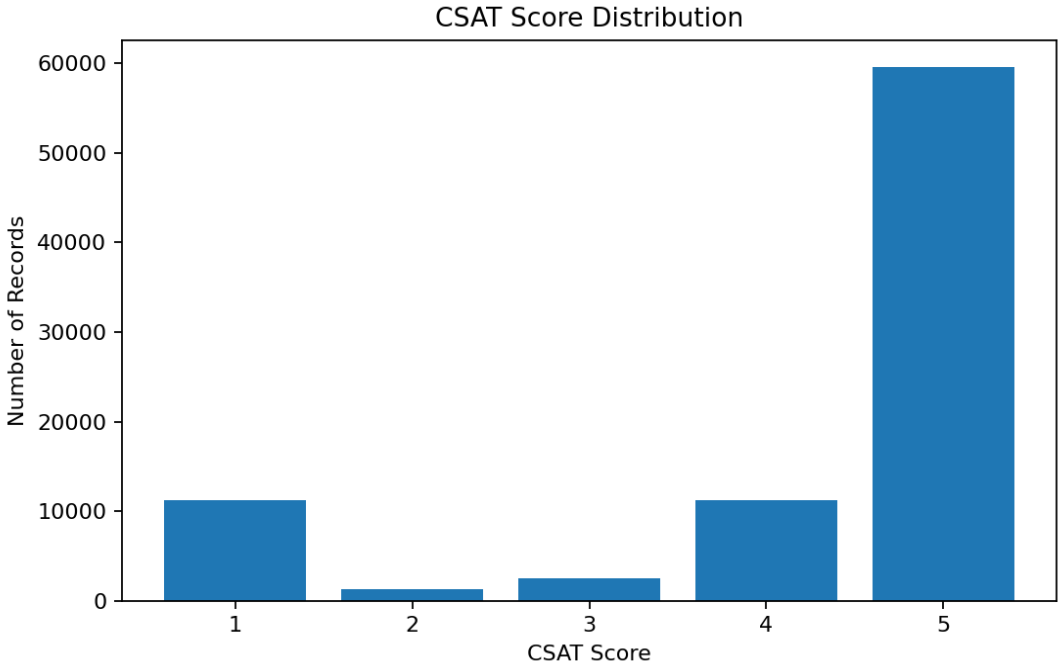
This project develops a feed-forward Artificial Neural Network to predict customer satisfaction scores from e-commerce customer-support interactions. The uploaded dataset contains 85,907 records and 20 columns.

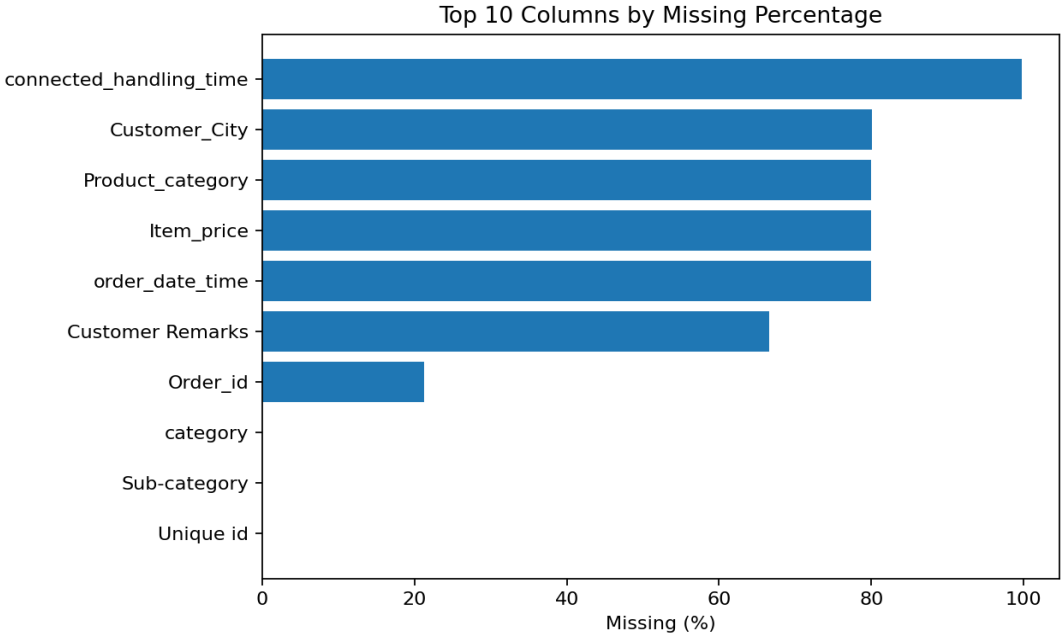
1. Business Problem

Customer satisfaction affects retention, repeat purchases and service quality. Predicting CSAT can help support teams identify interaction patterns associated with dissatisfaction and prioritize service improvements.

2. Dataset and EDA

The target contains five classes. CSAT 5 accounts for approximately 69.4% of observations, creating substantial class imbalance. Several order/product fields are missing in roughly 80% of records, while connected handling time is missing in almost all records.




Top 10 Columns by Missing Percentage
connected_handling_time
Customer_City
Product_category
Item_price
order_date_time
Customer Remarks
Order_id
category
Sub-category
Unique id
0
20
40
60
80
100
Missing (%)

3. Data Preparation and Feature Engineering

• Parse issue, response and order timestamps.

• Create response duration and order-to-issue duration.

• Create hour, weekday, weekend and circular time features.

• Create missingness indicators for sparse numerical fields.

• Create customer-remark length, word-count, punctuation and simple positive/negative lexicon features.

• One-hot encode categorical variables with unknown-category handling.

• Impute numerical values with medians and categorical values with the most frequent category.

• Apply Truncated SVD to produce 128 dense features for the ANN.

4. Leakage Prevention

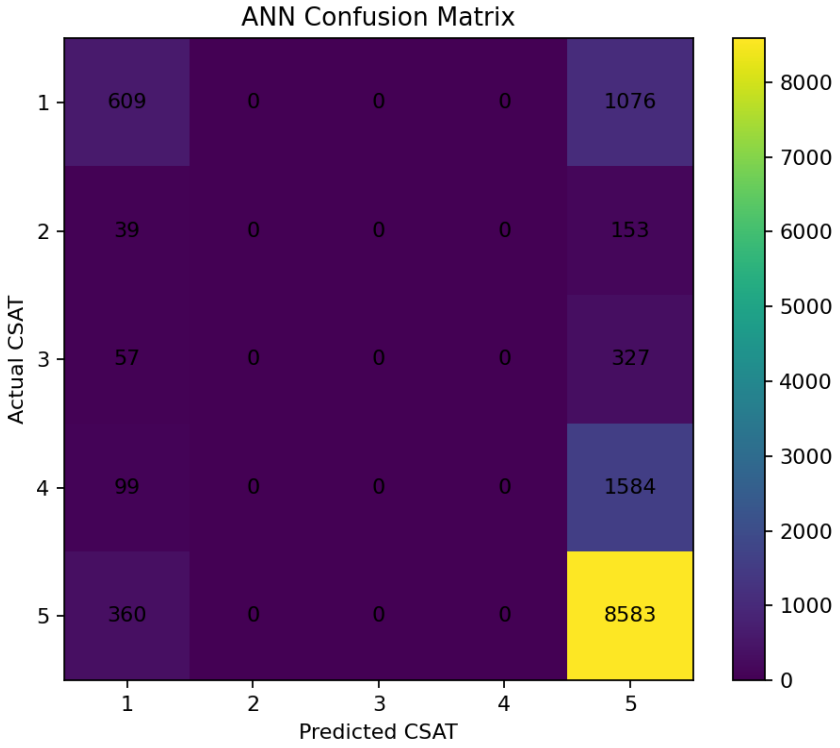
Survey response date is excluded because it occurs after the satisfaction outcome. Unique identifiers and order IDs are excluded. The project is explicitly framed as post-resolution prediction, so issue-response information and handling time are retained.

5. ANN Architecture

Input (128) → Dense 128 + BatchNorm + ReLU + Dropout(0.30) → Dense 64 + ReLU + Dropout(0.20) → Dense 32 + ReLU → Softmax output (5 classes). AdamW optimization and mild inverse-frequency class weighting are used.

6. Evaluation

Metric	Value
Test Accuracy	71.33%
Macro-F1	0.252
Weighted-F1	0.632
Majority Baseline Accuracy	69.40%
SVD Explained Variance	95.86%



7. Results and Interpretation

The model's test accuracy is above the majority-class baseline, but minority CSAT classes remain difficult to separate. This is visible in the confusion matrix and class-level report. Therefore, the system should not be described as a high-accuracy universal CSAT classifier. It is better viewed as a prototype for identifying broad satisfaction patterns and as a foundation for further improvement.

8. Local Deployment

A Streamlit application is included. Run `streamlit run app/app.py` after installing requirements. The app loads the saved preprocessing pipeline, SVD transformer and ANN weights, accepts a customer-support interaction, and displays the predicted CSAT score plus class probabilities.

9. Future Improvements

• Use a transformer-based text encoder for customer remarks.

• Collect more examples for CSAT 2–4.

• Evaluate ordinal classification because CSAT 1–5 is ordered.

• Compare ANN with CatBoost/XGBoost and calibrated baselines.

• Use cross-validation and systematic hyperparameter tuning.

• Monitor drift when deployed on future months of support data.